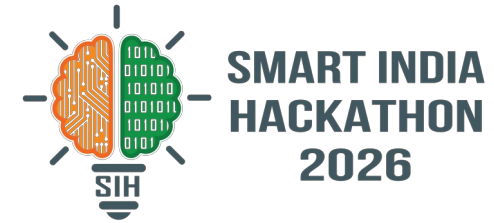
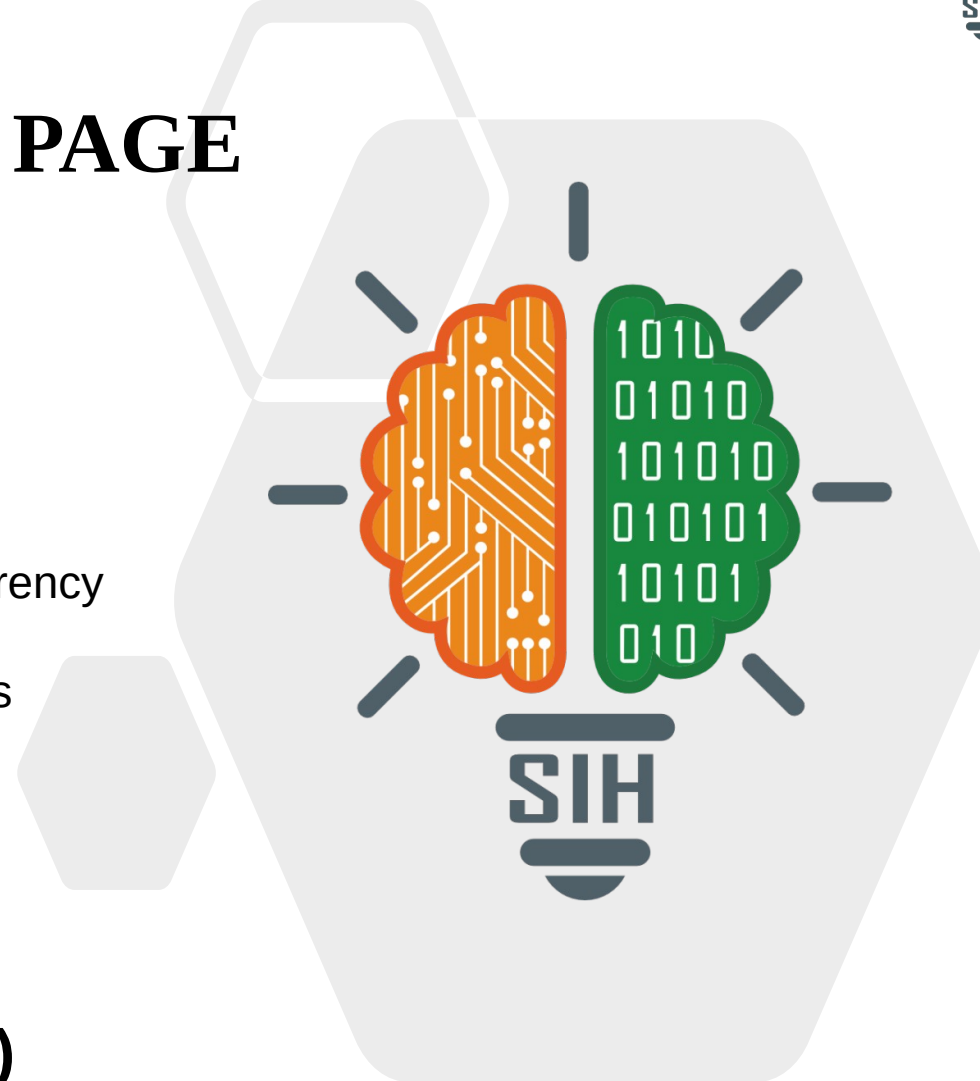


SMART INDIA HACKATHON 2026



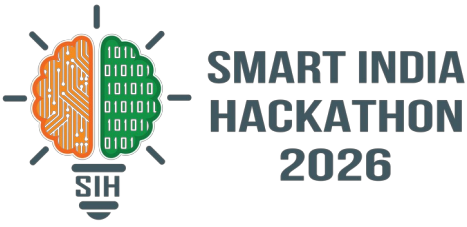
TITLE PAGE

- **Problem Statement ID** – SIH26183
- **Problem Statement Title-**
Real-Time Identification of Fraud-Linked Cryptocurrency
Exchanges from Victim-Reported Suspect Wallet
Addresses through Automated Blockchain Analytics
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-** 156766
- **Team Name (Registered on portal)**
COLD COFFEE



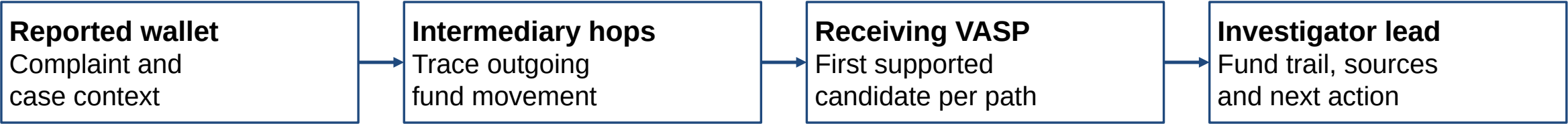


AEGIS FORENSICS



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation of the proposed solution



- How it addresses the problem

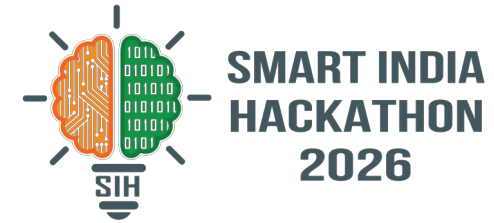
Continuous monitoring will alert investigators when funds move or a receiving VASP appears.
A case report will support review, preservation requests and coordination with the VASP.

- Innovation and uniqueness of the solution

Each lead links the transaction path to attribution sources and an investigative next step.
New transactions and revised labels will update findings while preserving earlier analysis.



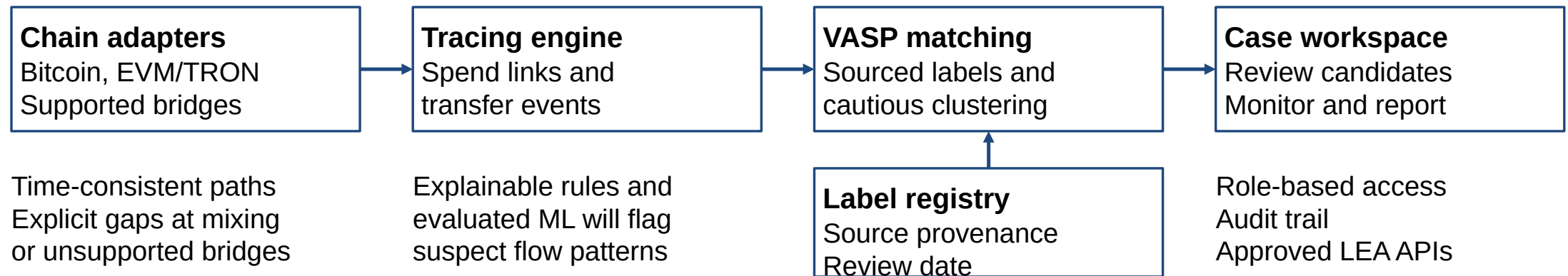
TECHNICAL APPROACH



- Technologies to be used (e.g. programming languages, frameworks, hardware)

React dashboard and FastAPI services, queued workers, incremental transaction indexing, persistent case storage and a versioned VASP-label registry.

- Methodology and process for implementation (Flow Charts/Images/ working prototype)



Findings will distinguish observed transactions, inferred links and unresolved attribution.

- Analysis of the feasibility of the idea



Label supply: reviewed public disclosures and authorized VASP data, with source history.

- Potential challenges and risks

Uncertain labels and flows

Provider limits and large histories

Sensitive cases and restricted access

- Strategies for overcoming these challenges

Version label sources, review clusters and show unresolved paths.

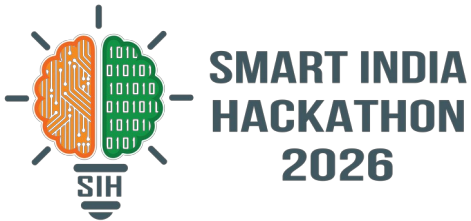
Index incrementally, cache queries and bound graph traversal.

Role-based permissions, audit logs and approved NCRP/SAHYOG APIs.

Validation: held-out labelled paths and hard negatives. Measure precision, coverage and alert latency.



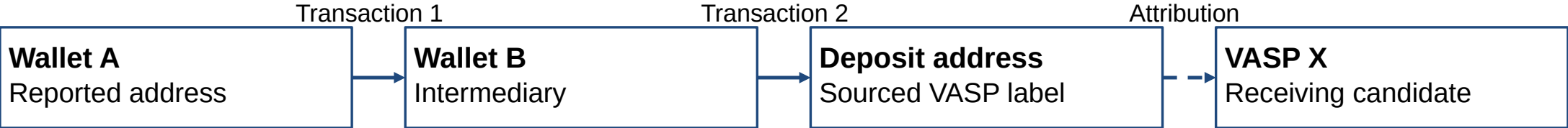
IMPACT AND BENEFITS



- Potential impact on the target audience

Investigators will receive a prioritized receiving-VASP lead with a reviewable fund trail.

Illustrative investigation



Finding Two-hop path, transaction IDs, amounts, timestamps and label source.

Next action Investigator reviews the lead and prepares a VASP preservation request.

Fictional illustration. Transaction paths alone do not prove exact fund continuity or wrongdoing.

- Benefits of the solution (social, economic, environmental, etc.)

Social

Support timely intervention

Economic

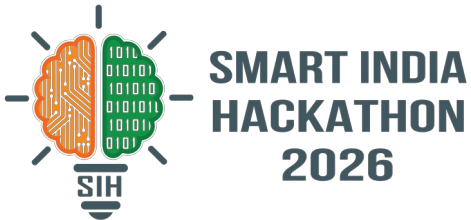
Reduce repeated tracing work

Operational

Standardize VASP coordination



RESEARCH AND REFERENCES



- Details / Links of the reference and research work

Address clustering	Meiklejohn et al. (2013), A Fistful of Bitcoins: Characterizing Payments Among Men with No Names. ACM IMC paper
ML-assisted analysis	Weber et al. (2019), Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics. arXiv:1908.02591
Blockchain data	Blockstream Esplora: address and transaction API documentation. Official API documentation
Attribution sources	Binance, Our Commitment to Transparency (2022): published wallet disclosures. Historical disclosure; not a deposit-account registry
AML context	FATF (2021), Updated Guidance for a Risk-Based Approach to Virtual Assets and VASPs. FATF guidance
Evidence handling	NIST SP 800-86 (2006), Guide to Integrating Forensic Techniques into Incident Response. NIST guidance; not a guarantee of legal admissibility